

POM 610-631:

Supply Chain Management I

Polaris Case Presentation

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Global Supply Chain Network

US Assembly vs Overseas Assembly

Reasons to only assembly in the US

- American Manufacturing
 - Both customers and employees are proud that the product is manufactured in the United States
- Supply Chain Visibility
 - Order communication, production and transportation (fuel price volatility)
- Product Returns
 - Quality (State of the Art Injection Molding Plant at Roseau)

Reasons to only assembly in the US

- Higher Hidden Cost
 - Managing Exchange Rate Risks
 - Invoice Errors
- Tariffs and Taxes
 - 5% production tariffs for China Plan
- Freight Costs
 - Transportation modes and quantity



NPV Calculations

US vs Mexico vs China



ABOVE: US Plant; NPV Calculations (\$53,058,088.51) for Years 2010 - 2015

ABOVE: US Plant; Projected Production, Labor, and Transportation Costs for 2010 - 2015



NPV Calculation: Mexico Plant

	YEAR						
COSTS (in MXN)	2010	2011	2012	2013	2014	2015	TOTAL COSTS
Startup Costs plus Severance Package	132,920,750.00	0.00	0.00	0.00	0.00	0.00	132,920,750.00
Production Costs	0.00	69,568,934.89	69,672,402.94	69,778,975.03	69,888,744.28	70,001,806.61	348,910,863.74
Tariffs	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Transportation	0.00	22,640,254.70	22,640,254.70	22,640,254.70	22,640,254.70	22,640,254.70	113,201,273.50
TOTAL Annual COSTS (in MXN)	132,920,750.00	92,209,189.59	92,312,657.64	92,419,229.73	92,528,998.98	92,642,061.31	595,032,887.24
TOTAL ANNUAL COSTS (in USD)	\$10,700,000.00	\$ 7,422,756.26	\$ 7,431,085.34	\$ 7,439,664.30	\$ 7,448,500.62	\$ 7,457,602.04	\$47,899,608.55

ABOVE: Mexico Plant; Projected Production, Labor, and Transportation Costs for 2010 - 2015

Sum of 2010 - 2015 NPV's	
(in MXN)	(in USD)
483,196,656.36	\$ 38,896,893.25

ABOVE: Mexico Plant; NPV Calculations (\$38,896,893.25) for Years 2010 - 2015

YEAR						
2015 NPV	2010	2011	2012	2013	2014	2015
						92,642,061.31
					84,220,055.73	\$ 7,457,602.04
			69,603,351.85	\$ 6,163,307.47	76,563,687.03	\$ 6,779,638.22
		63,275,774.40	\$ 5,603,006.79			
NPV (in MXN)	57,523,431.28	\$ 5,093,642.54				
NPV (in USD)	\$ 4,630,584.12					
YEAR						
2014 NPV	2010	2011	2012	2013	2014	2015
					92,528,998.98	
				84,117,271.80	\$ 7,448,500.62	
			76,470,247.09	\$ 6,771,364.20		
		69,518,406.44	\$ 6,155,785.64			
NPV (in MXN)	63,198,551.31	\$ 5,596,168.76				
NPV (in USD)	\$ 5,087,426.15					
YEAR						
2013 NPV	2010	2011	2012	2013	2014	2015
				92,419,229.73		
			84,017,481.57	\$ 7,439,664.30		
		76,379,528.70	\$ 6,763,331.18			
		69,435,935.18	\$ 6,148,482.89			
NPV (in MXN)	69,435,935.18	\$ 6,148,482.89				
NPV (in USD)	\$ 5,589,529.90					
YEAR						
2012 NPV	2010	2011	2012	2013	2014	2015
			92,312,657.64			
		83,920,597.85	\$ 7,431,085.34			
NPV (in MXN)	76,291,452.59	\$ 6,755,532.13				
NPV (in USD)	\$ 6,141,392.84					
YEAR						
2011 NPV	2010	2011	2012	2013	2014	2015
		92,209,189.59				
NPV (in MXN)	83,826,535.99	\$ 7,422,756.26				
NPV (in USD)	\$ 6,747,960.23					
YEAR						
2010 NPV	2010	2011	2012	2013	2014	2015
NPV (in MXN)	132,920,750.00					
NPV (in USD)	\$ 10,700,000.00					



NPV Calculation: China Plant

	YEAR						
COSTS (in CNY)	2010	2011	2012	2013	2014	2015	TOTAL COSTS
Startup Costs plus Severance Package	76,160,000.00	0.00	0.00	0.00	0.00	0.00	76,160,000.00
Production Costs	0.00	30,482,580.48	30,813,717.55	31,194,525.18	31,632,453.96	32,136,072.06	156,259,349.24
Tariffs	0.00	2,460,829.02	2,477,385.88	2,496,426.26	2,518,322.70	2,543,503.60	12,496,467.46
Transportation	0.00	18,734,000.00	18,734,000.00	18,734,000.00	18,734,000.00	18,734,000.00	93,670,000.00
TOTAL Annual COSTS (in CNY)	76,160,000.00	51,677,409.50	52,025,103.43	52,424,951.44	52,884,776.66	53,413,575.66	338,585,816.70
TOTAL ANNUAL COSTS (in USD)	\$ 11,200,000.00	\$ 7,599,619.04	\$ 7,650,750.50	\$ 7,709,551.68	\$ 7,777,173.04	\$ 7,854,937.60	\$ 49,792,031.87

ABOVE: China Plant; Projected Production, Labor, and Transportation Costs for 2010 - 2015

Sum of 2010 - 2015 NPV's	
(in CNY)	(in USD)
274,809,700.50	\$40,413,191.25

ABOVE: China Plant; NPV Calculations (\$40,413,191.25) for Years 2010 - 2015

YEAR						
2015 NPV	2010	2011	2012	2013	2014	2015
						53,413,575.66
					48,557,796.05	\$7,854,937.60
				44,143,450.96	\$7,140,852.36	
			40,130,409.96	\$6,491,683.96		
		36,482,190.87	\$5,901,530.88			
NPV (in CNY)	33,165,628.07	\$5,365,028.07				
NPV (in USD)	\$ 4,877,298.25					

YEAR						
2014 NPV	2010	2011	2012	2013	2014	2015
					52,884,776.66	
					48,077,069.69	\$7,777,173.04
				43,706,426.99	\$7,070,157.31	
			39,733,115.45	\$6,427,415.73		
NPV (in CNY)	36,121,014.04	\$5,843,105.21				
NPV (in USD)	\$ 5,311,913.83					

YEAR						
2013 NPV	2010	2011	2012	2013	2014	2015
					52,424,951.44	
					47,659,046.77	\$7,709,551.68
				43,326,406.15	\$7,008,683.35	
NPV (in CNY)	39,387,641.96	\$6,371,530.32				
NPV (in USD)	\$ 5,792,300.29					

YEAR						
2012 NPV	2010	2011	2012	2013	2014	2015
NPV (in CNY)	42,995,953.25	\$6,955,227.73				
NPV (in USD)	\$ 6,322,934.30					

YEAR						
2011 NPV	2010	2011	2012	2013	2014	2015
NPV (in CNY)	46,979,463.19	\$7,599,619.04				
NPV (in USD)	\$ 6,908,744.59					

YEAR						
2010 NPV	2010	2011	2012	2013	2014	2015
NPV (in CNY)	76,160,000.00					
NPV (in USD)	\$11,200,000.00					

Sensitivity Analysis

Mexico vs China



Sensitivity Analysis: Mexico Plant

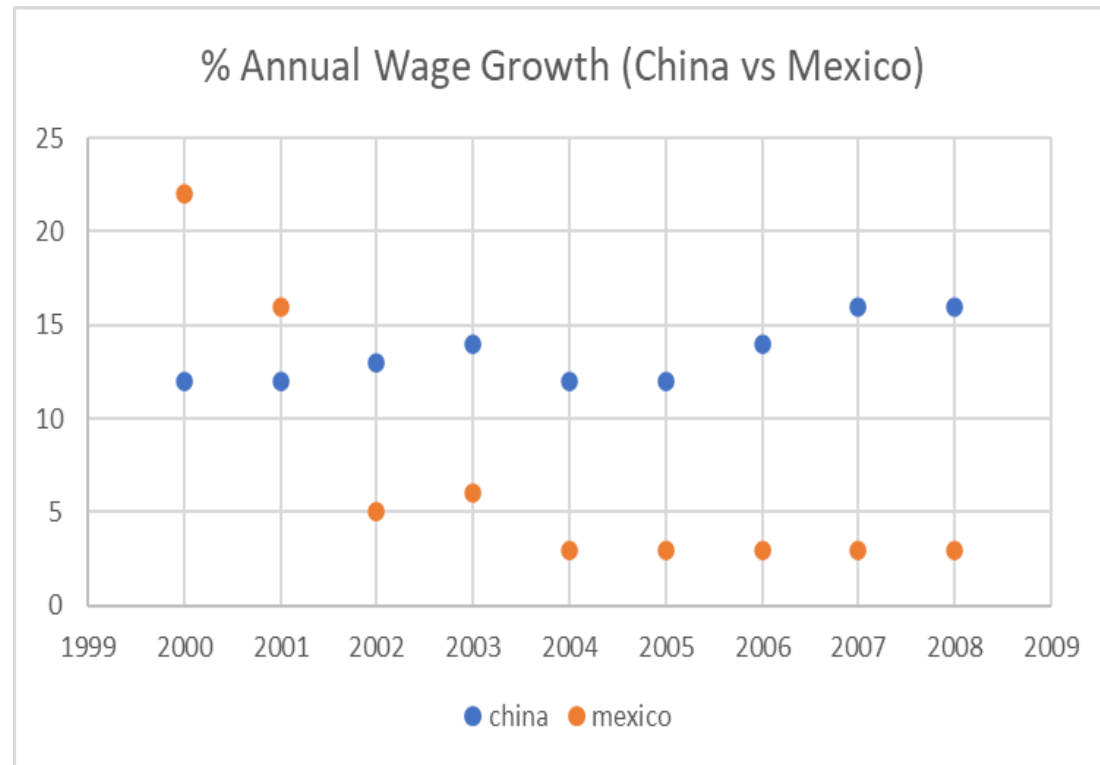
SENSITIVITY ANALYSIS: MEXICO							
			EXCHANGE RATE				
		NPV (below)	↑ 25%	↑ 10%	No Δ	↓ 10%	↓ 25%
		\$ 38,896,893.25	15.52	13.66	12.42	11.18	9.32
2011 LABOR COST	↑ 25%	\$ 71,853.00	\$ 34,698,693.70	\$ 37,022,881.29	\$ 38,964,290.95	\$ 41,337,124.98	\$ 46,082,793.03
	↑ 10%	\$ 63,230.00	\$ 34,668,387.93	\$ 36,988,454.01	\$ 38,926,420.94	\$ 41,295,047.19	\$ 46,032,299.69
	No Δ	\$ 57,482.25	\$ 34,648,187.31	\$ 36,965,506.16	\$ 38,901,178.30	\$ 41,266,999.81	\$ 45,998,642.83
	↓ 10%	\$ 51,734.00	\$ 34,627,984.93	\$ 36,942,556.30	\$ 38,875,933.46	\$ 41,238,949.99	\$ 45,964,983.05
	↓ 25%	\$ 43,111.00	\$ 34,597,679.17	\$ 36,908,129.02	\$ 38,838,063.46	\$ 41,196,872.21	\$ 45,914,489.71
Best Case VS Worst Case Δ: \$11,485,113.86 (33.2%)							



Sensitivity Analysis: China Plant

SENSITIVITY ANALYSIS: CHINA								
					EXCHANGE RATE			
			NPV (below)	↑ 25%	↑ 10%	No Δ	↓ 10%	↓ 25%
			\$ 40,413,191.25	8.5	7.48	6.8	6.12	5.1
2011 LABOR COST	↑ 25%	\$ 45,991.00	\$36,825,688.46	\$ 38,824,764.37	\$40,490,660.97	\$42,526,756.81	\$46,598,948.48	
	↑ 10%	\$ 40,472.00	\$36,788,501.62	\$ 38,782,506.60	\$40,444,177.41	\$42,475,108.41	\$46,536,970.41	
	No Δ	\$ 36,793.01	\$36,763,712.70	\$ 38,754,337.37	\$40,413,191.27	\$42,440,679.36	\$46,495,655.55	
	↓ 10%	\$ 33,113.00	\$36,738,916.91	\$ 38,726,160.34	\$40,382,196.53	\$42,406,240.76	\$46,454,329.23	
	↓ 25%	\$ 27,236.39	\$36,699,320.50	\$ 38,681,164.42	\$40,332,701.02	\$42,351,245.75	\$46,388,335.22	
Best Case VS Worst Case Δ: \$9,899,627.98 (26%)								

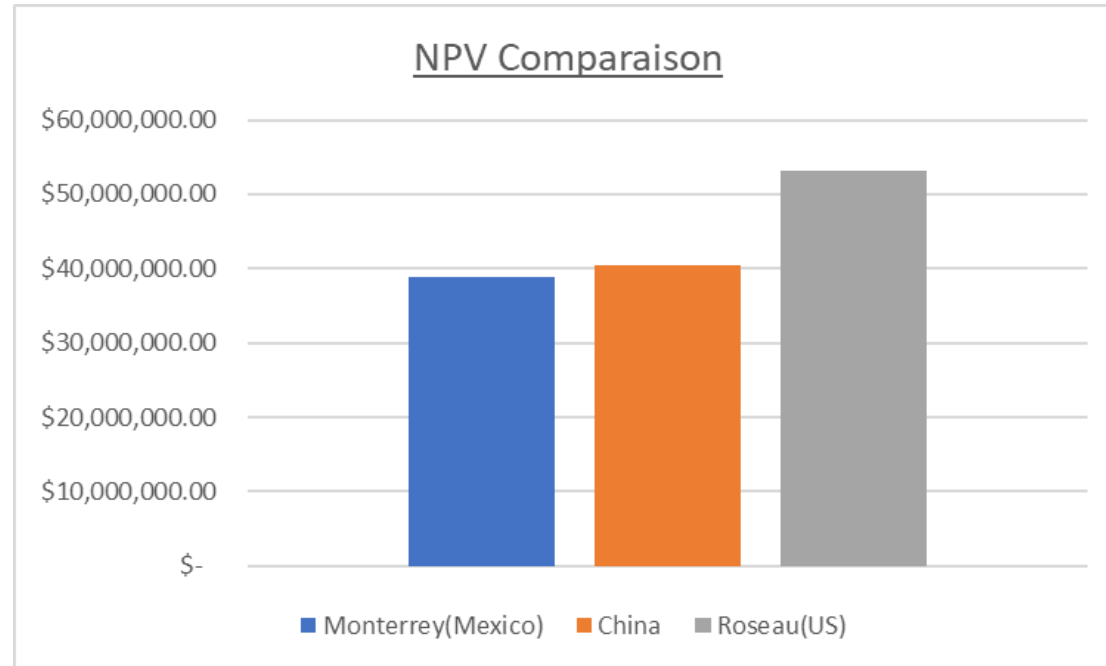
Manufacturing Location Decision (Beside cost)



Responsiveness of the Supply Chain

- Procurement Risk
 - Exchange rate risk
 - Significant part of the business dependent of a single external source
- Systems Risk
 - Information System Failure
- Disruptions
 - Covid 19 example¹³

Flexibility of the Supply Chain



- Receivable Risk
 - Customer quantity (change in demand)
- Capacity Risk
 - Cost of Capacity
 - Capacity Flexibility
- Forecast Risk
 - Incorrect forecast(production delays)

ANY
QUESTIONS
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